

Automotive Ethernet Bus Capture Measurement Plugin for Instrument Studio in LabVIEW

This plugin is an easy-to-use Automotive Ethernet Bus capture, with sorting and filtering display options. It uses the Measurement Plugin SDK.

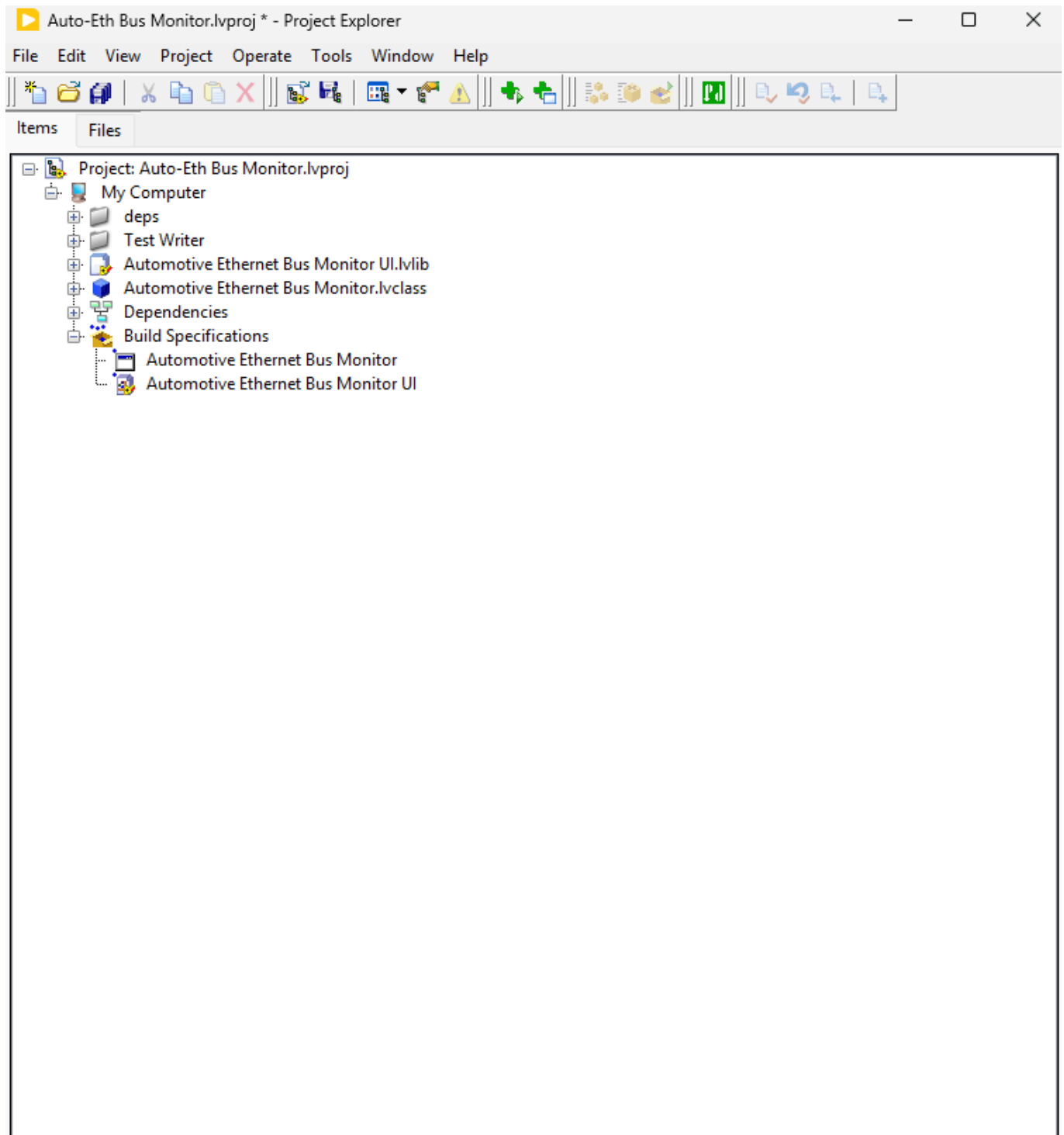
Contents

1. The service logic main file is the **Automotive Ethernet Bus Monitor\Measurement Logic.vi**.
2. The UI part is implemented in the **Automotive Ethernet Bus Monitor UI\Measurement UI.vi** file.

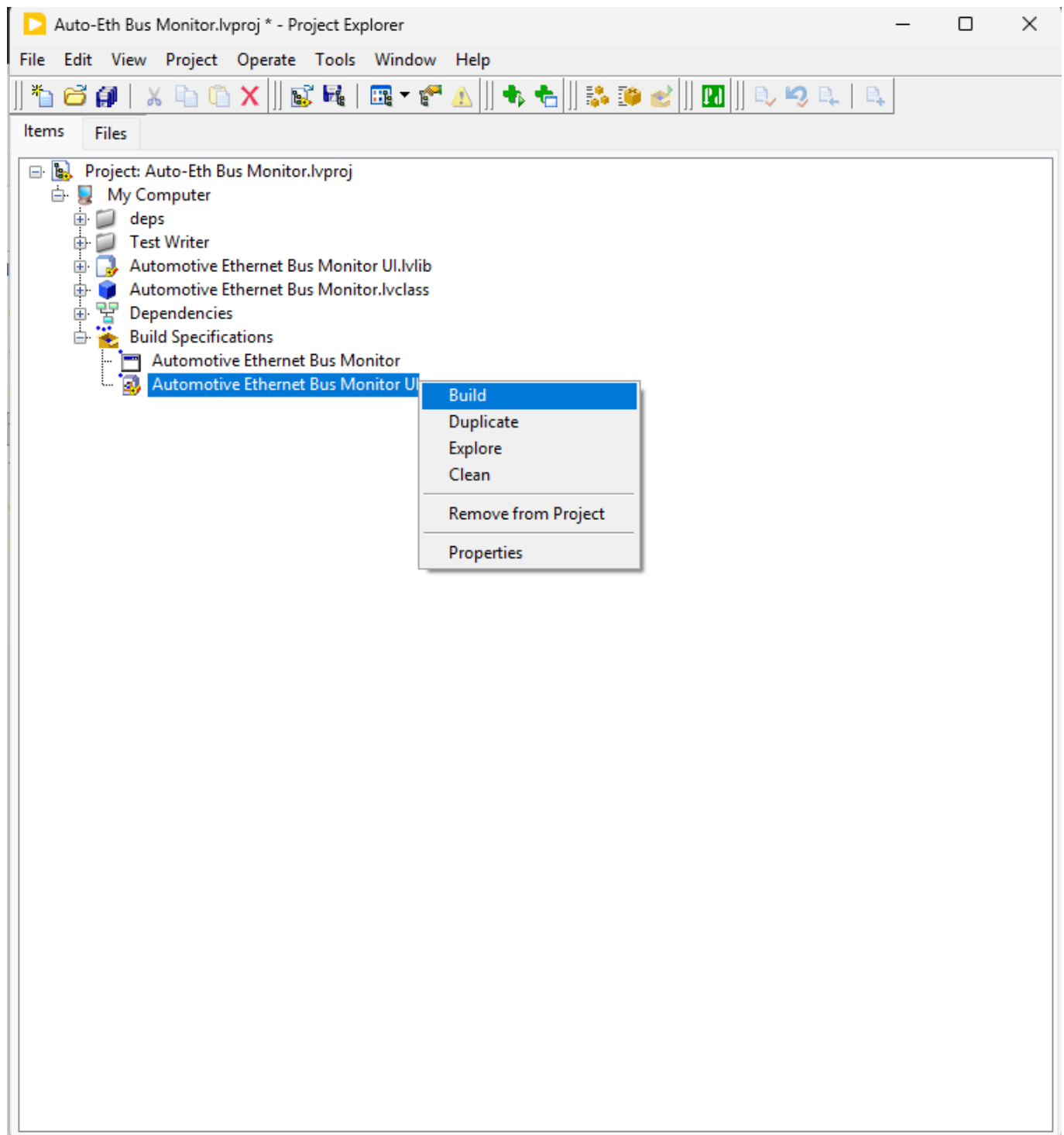
The plugin uses the **Tables and Filtering** library for sorting and filtering, and the OOB (Out-Of-Band) gRPC Service for the interactive UI part (Pause / Resume).

How to run this plugin

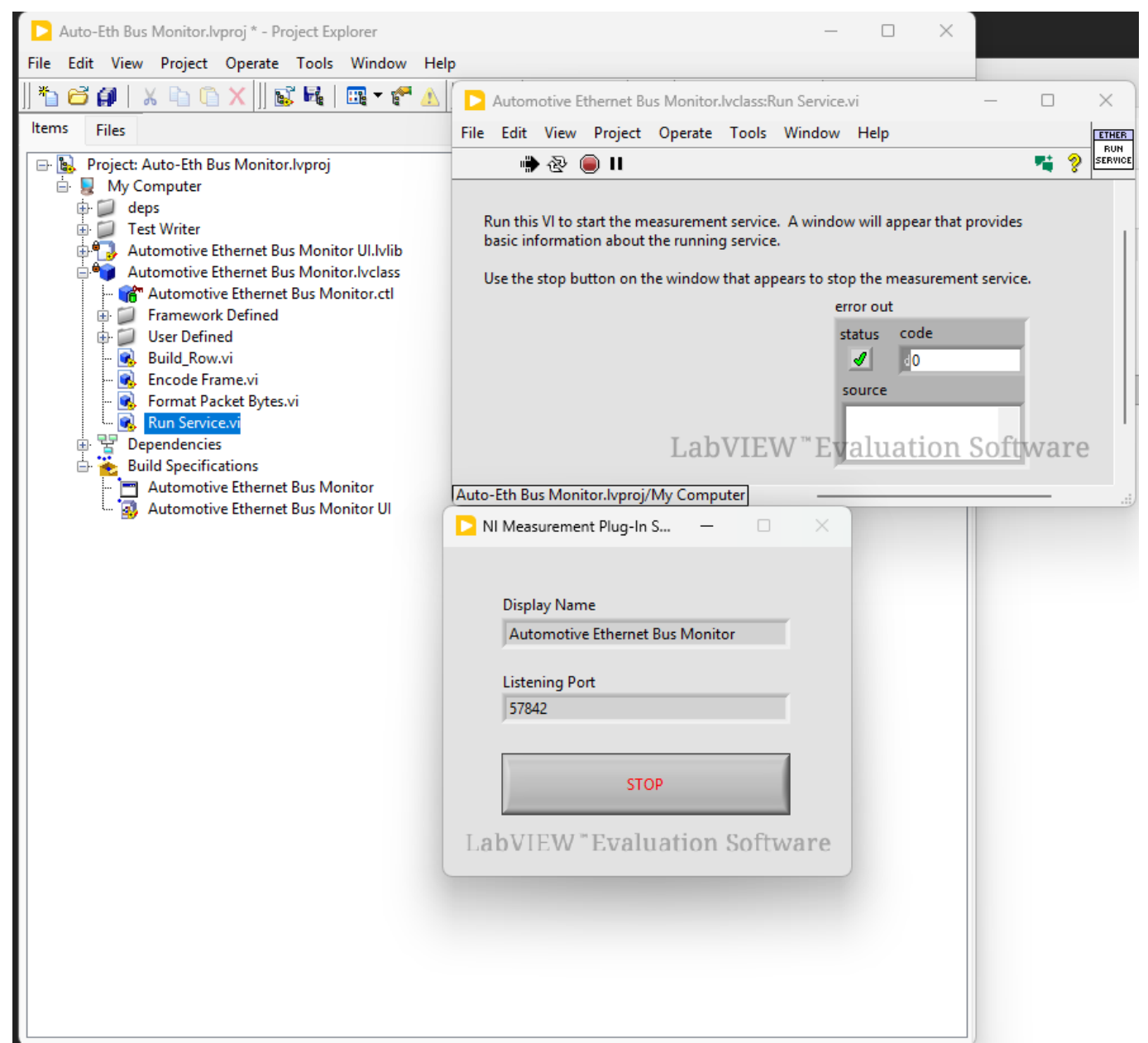
Open the Auto-Eth Bus Monitor.lvproj in LabVIEW 2025.



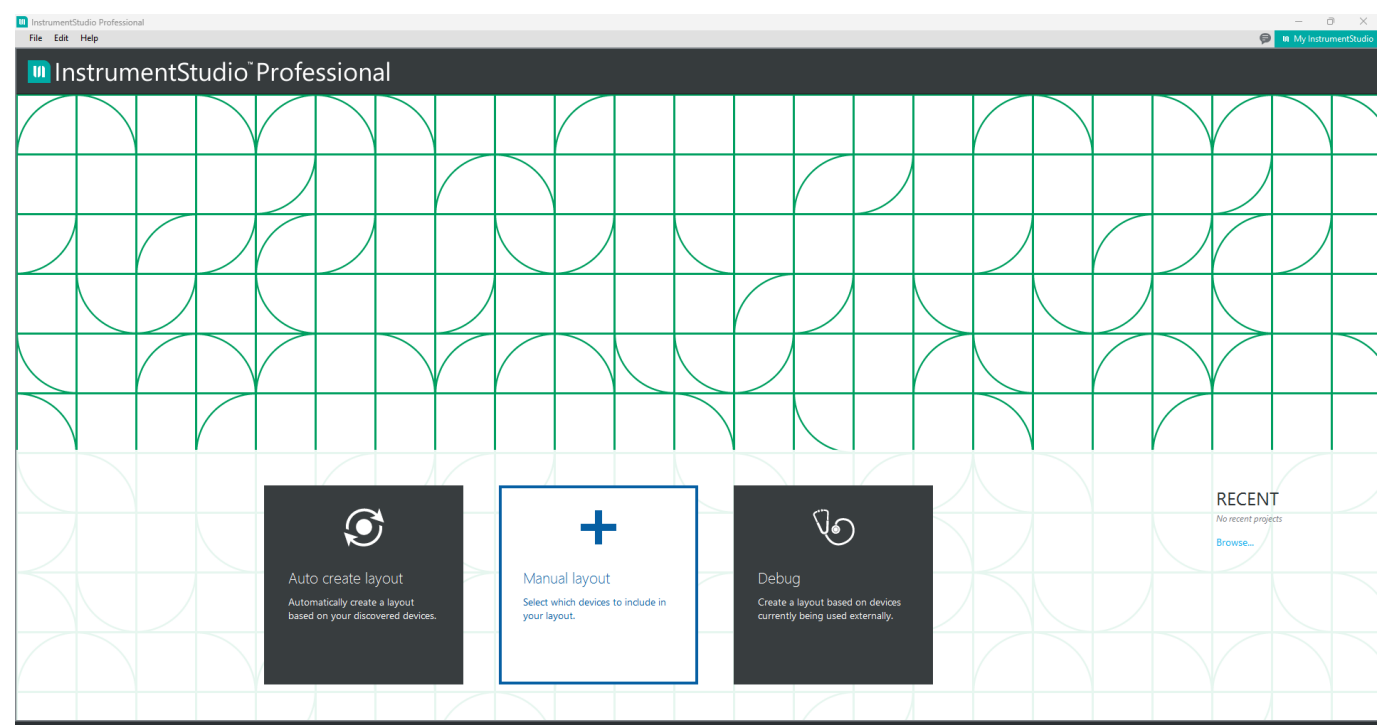
Build both the UI and the exe inside the project by right-clicking and selecting build on both of them.



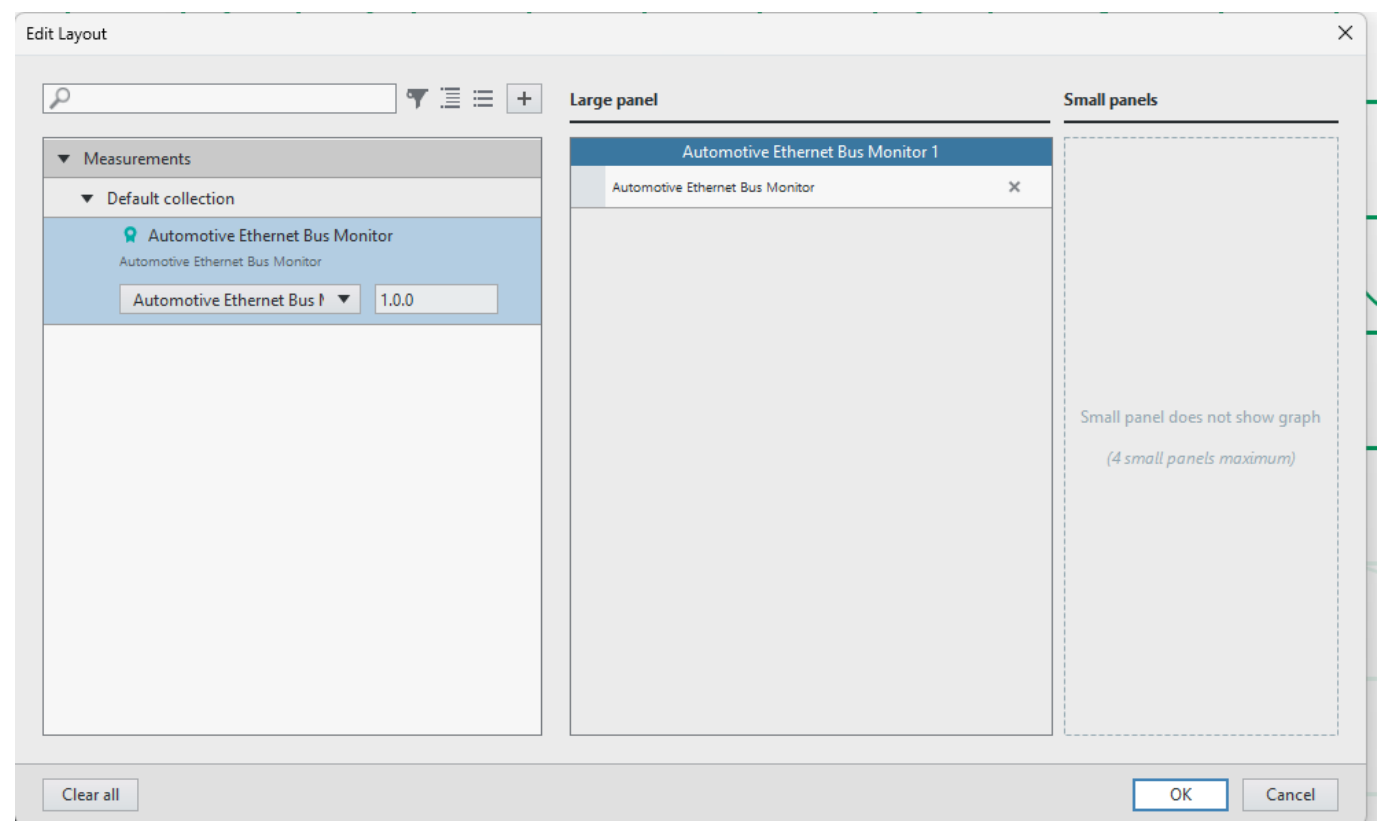
Open up the Run Service VI and run the plugin. This will create the measurement service.



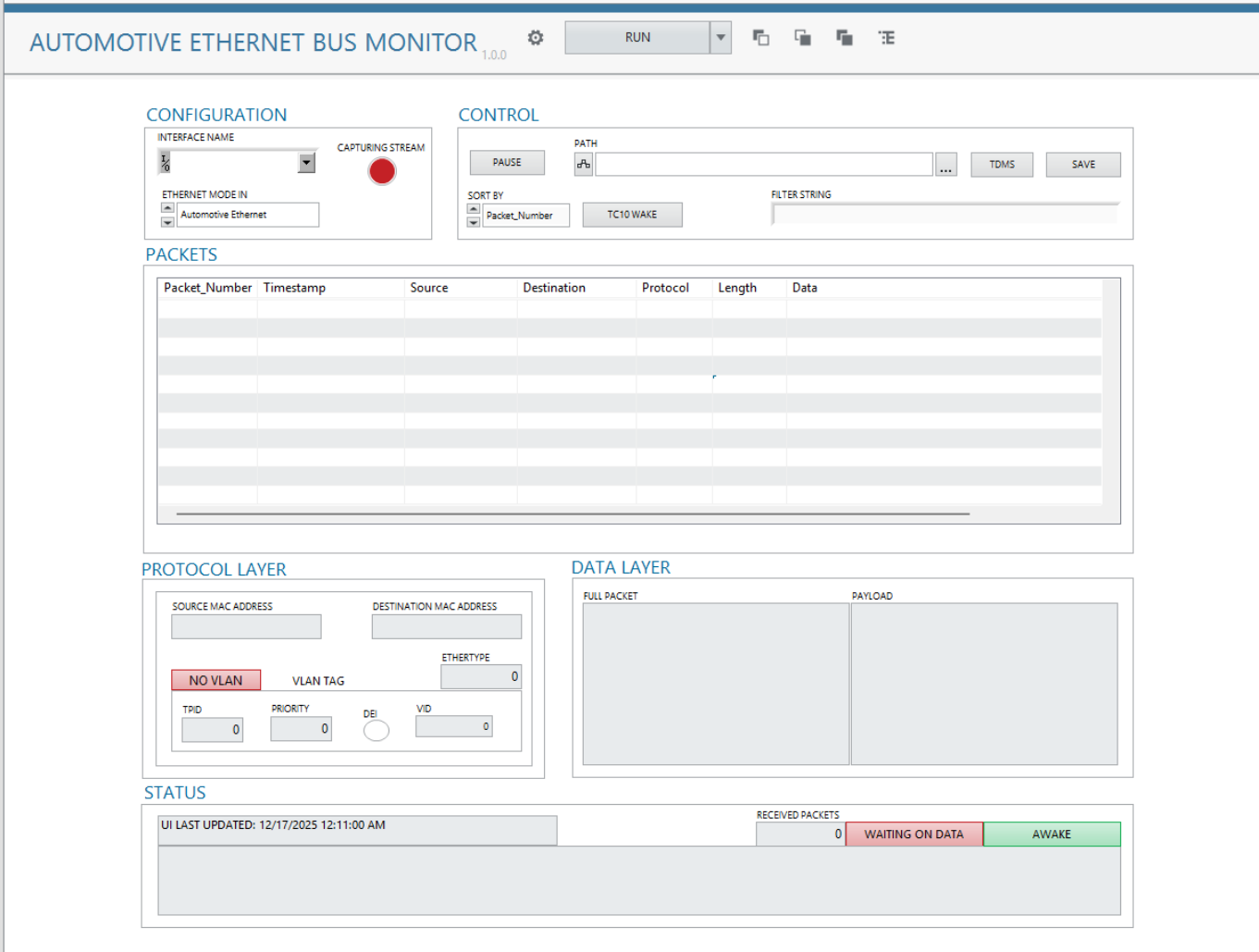
Once the service is running, open up Measurement Studio and select manual layout (pay attention to the monitor resolution; a minimum 1920 x 1080 resolution should be used).



Please select the service, and in the drop-down menu, select 'Create large panel'.



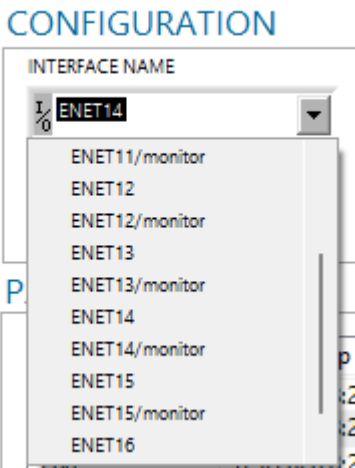
The opened UI should look like this:



Please see troubleshooting if it does not look like the picture above.

How to configure the plugin

First, select the Automotive Ethernet Interface from the drop down menu of available XNET interfaces. The user can select their preferred Interface or their preferred Interface/monitor. If it is not selected, the monitor keyword is added anyway.



Set the ETHERNET MODE IN to Automotive Ethernet. Support is not yet available for Standard Ethernet.

Configuration

Interface Name In

ENET9

Capturing Stream



ETHERNET MODE IN

⬆

Automotive Ethernet

⬇

Press the Run Button. When the first packet is received, the 'Capturing Stream' icon should light up.

How to control the plugin

Once the plugin starts running, it should look similar to this.

CONFIGURATION

INTERFACE NAME

1/20 ENET14

CAPTURING STREAM



ETHERNET MODE IN

⬆Automotive Ethernet⬇

CONTROL

RESUME

PATH

⌵

...

SAVE FORMAT

⬆TDMS⬇

SAVE

SORT BY

⬆Packet_Number⬇

TC10 SLEEP

FILTER STRING

PACKETS

Packet_Number	Timestamp	Source	Destination	Protocol	Length	Data
99	1/6/2026 4:46:48 PM	00:80:2F:42:A0:49	01:00:5E:00:00:FB	800	113	01 00 5E 00 00 FB 00 80 2F 42 A0 49 08 00 45 00 00 63 6
100	1/6/2026 4:46:48 PM	00:80:2F:42:A0:49	FF:FF:FF:FF:FF:FF	806	60	FF FF FF FF FF FF 00 80 2F 42 A0 49 08 06 00 01 08 00 0
101	1/6/2026 4:46:48 PM	00:80:2F:42:A0:49	01:00:5E:00:00:16	800	60	01 00 5E 00 00 16 00 80 2F 42 A0 49 08 00 46 00 00 28 6
102	1/6/2026 4:46:48 PM	00:80:2F:42:A0:4A	FF:FF:FF:FF:FF:FF	806	60	FF FF FF FF FF FF 00 80 2F 42 A0 4A 08 06 00 01 08 00 0
103	1/6/2026 4:46:48 PM	00:80:2F:42:A0:4A	01:00:5E:00:00:16	800	60	01 00 5E 00 00 16 00 80 2F 42 A0 4A 08 00 46 00 00 28 2
104	1/6/2026 4:46:48 PM	00:80:2F:42:A0:49	33:33:00:00:00:16	86DD	90	33 33 00 00 00 16 00 80 2F 42 A0 49 86 DD 60 00 00 00 0
105	1/6/2026 4:46:48 PM	00:80:2F:42:A0:4A	33:33:00:00:00:16	86DD	90	33 33 00 00 00 16 00 80 2F 42 A0 4A 86 DD 60 00 00 00 0
106	1/6/2026 4:46:48 PM	00:80:2F:42:A0:49	01:00:5E:00:00:FB	800	226	01 00 5E 00 00 FB 00 80 2F 42 A0 49 08 00 45 00 00 D4 6
107	1/6/2026 4:46:48 PM	00:80:2F:42:A0:4A	01:00:5E:00:00:FB	800	226	01 00 5E 00 00 FB 00 80 2F 42 A0 4A 08 00 45 00 00 D4 :
108	1/6/2026 4:46:48 PM	00:80:2F:42:A0:49	01:00:5E:00:00:FB	800	257	01 00 5E 00 00 FB 00 80 2F 42 A0 49 08 00 45 00 00 F3 6
109	1/6/2026 4:46:48 PM	00:80:2F:42:A0:4A	01:00:5E:00:00:FB	800	256	01 00 5E 00 00 FB 00 80 2F 42 A0 4A 08 00 45 00 00 F2 :
110	1/6/2026 4:46:48 PM	00:80:2F:42:A0:49	01:00:5E:00:00:FB	800	292	01 00 5E 00 00 FB 00 80 2F 42 A0 49 08 00 45 00 01 16 6

PROTOCOL LAYER

SOURCE MAC ADDRESS

00:80:2F:42:A0:4A

DESTINATION MAC ADDRESS

01:00:5E:00:00:16

ETHERTYPE

0

NO VLAN

VLAN TAG

TPID

0

PRIORITY

0

DEI

VID

0

DATA LAYER

FULL PACKET

01 00 5E 00 00 16 00 80 2F 42 A0 4A 08 00 46 00 00 28 2F 3B 00 00 01 02 84 76 A9 FE E7 09 E0 00 00 16 94 04 00 00 22 00 F9 01 00 00 00 01 04 00 00 00 E0 00 00 FC 00 00 00 00 00 00

PAYLOAD

F (/; „v@pç à □" " ù□ □ à ù

STATUS

UI LAST UPDATED: 1/6/2026 4:48:53 PM

RECEIVED PACKETS

245

WAITING ON DATA

AWAKE

Paused Stream

Packets should be collected and displayed in the table as they come in through the bus.

PACKETS

Packet_Number	Timestamp	Source	Destination	Protocol	Length	Data
99	1/6/2026 4:46:48 PM	00:80:2F:42:A0:49	01:00:5E:00:00:FB	800	113	01 00 5E 00 00 FB 00 80 2F 42 A0 49 08 00 45 00 00 63 6
100	1/6/2026 4:46:48 PM	00:80:2F:42:A0:49	FF:FF:FF:FF:FF:FF	806	60	FF FF FF FF FF FF 00 80 2F 42 A0 49 08 06 00 01 08 00 0
101	1/6/2026 4:46:48 PM	00:80:2F:42:A0:49	01:00:5E:00:00:16	800	60	01 00 5E 00 00 16 00 80 2F 42 A0 49 08 00 46 00 00 28 6
102	1/6/2026 4:46:48 PM	00:80:2F:42:A0:4A	FF:FF:FF:FF:FF:FF	806	60	FF FF FF FF FF FF 00 80 2F 42 A0 4A 08 06 00 01 08 00 0
103	1/6/2026 4:46:48 PM	00:80:2F:42:A0:4A	01:00:5E:00:00:16	800	60	01 00 5E 00 00 16 00 80 2F 42 A0 4A 08 00 46 00 00 28 2
104	1/6/2026 4:46:48 PM	00:80:2F:42:A0:49	33:33:00:00:00:16	86DD	90	33 33 00 00 00 16 00 80 2F 42 A0 49 86 DD 60 00 00 00 0
105	1/6/2026 4:46:48 PM	00:80:2F:42:A0:4A	33:33:00:00:00:16	86DD	90	33 33 00 00 00 16 00 80 2F 42 A0 4A 86 DD 60 00 00 00 0
106	1/6/2026 4:46:48 PM	00:80:2F:42:A0:49	01:00:5E:00:00:FB	800	226	01 00 5E 00 00 FB 00 80 2F 42 A0 49 08 00 45 00 00 D4 6
107	1/6/2026 4:46:48 PM	00:80:2F:42:A0:4A	01:00:5E:00:00:FB	800	226	01 00 5E 00 00 FB 00 80 2F 42 A0 4A 08 00 45 00 00 D4 :
108	1/6/2026 4:46:48 PM	00:80:2F:42:A0:49	01:00:5E:00:00:FB	800	257	01 00 5E 00 00 FB 00 80 2F 42 A0 49 08 00 45 00 00 F3 6
109	1/6/2026 4:46:48 PM	00:80:2F:42:A0:4A	01:00:5E:00:00:FB	800	256	01 00 5E 00 00 FB 00 80 2F 42 A0 4A 08 00 45 00 00 F2 3
110	1/6/2026 4:46:48 PM	00:80:2F:42:A0:49	01:00:5E:00:00:FB	800	292	01 00 5E 00 00 FB 00 80 2F 42 A0 49 08 00 45 00 01 16 6

Scrollbars exist on the sides of the table, so the user can scroll vertically or horizontally.

The control panel is displayed at the top of the UI.

This control panel allows the user to do the following:

- 1. Pause/Resume the data stream.
- 2. Save the table output to a CSV.
- 3. Sort the table by the column headers.
- 4. [Filter](#) the table using boolean logic.

Some additional tools are built in to help the user understand the state of the plugin.

Inside the control panel, pressing the 'Pause' button will cause the color and text to change, indicating the plugin is paused. Likewise, pressing the 'Resume' button will unpause the plugin. When the plugin is paused, the packets will still be collected, but the UI will not display the new packets until the system is unpause

CONTROL

PAUSE

PATH

...

TDMS

SAVE

SORT BY

Packet_Number

TC10 SLEEP

FILTER STRING

The Status field at the bottom of the UI displays the latest status reported by the plugin. This will indicate if the stream is paused, when a file is saved to disk, connection status, or when an error occurs in the plugin. The Packets Received field also displays the number of packets received on the bus. The Displaying Icon is displayed in the status window when new data is getting pulled from the bus. If there is no new data, "Waiting For Data" is displayed instead.

STATUS

UI LAST UPDATED: 12/16/2025 11:56:12 PM

RECEIVED PACKETS

229

DISPLAYING

AWAKE

OOB gRPC Connected!

The packet table will continually display the latest data, auto scrolling so the latest data is at the bottom. Pausing the plugin will allow the user to scroll back through the collected data.

When filtering the data, the table will only display the filtered results, as such the table will auto scroll to the top of the filtered results.

TC10 Functionality

The TC10 Wake and TC10 Sleep button will set the power mode for the interface. This only works for devices that are TC10 enabled. The "Sleeping/Wake Field" in the bottom right exists to indicate to the user the TC10 status. Devices that are not TC10 enabled will have the default high power state and will indicate 'Wake'. Sometimes, the interface is asleep before the user starts the plugin, in which case the TC10 Wake button must be hit to allow the packets to start flowing.

STATUS

UI LAST UPDATED: 1/5/2026 9:50:52 PM

RECEIVED PACKETS0

WAITING ON DATA

AWAKE

Waking Up ENET14

STATUS

UI LAST UPDATED: 1/5/2026 9:50:52 PM

RECEIVED PACKETS0

WAITING ON DATA

SLEEPING

Setting ENET14 to sleep

If the device is not TC10 Enabled, an error message indicating as such will appear in the status window.

STATUS

UI LAST UPDATED: 1/5/2026 9:52:10 PM

RECEIVED PACKETS14

DISPLAYING

AWAKE

The sleep capability is either disabled or not available for this device or interface configuration.

There is not a running list of NI devices that are TC10 enabled. The best way to find out if the device is TC10 Enabled is to to check the manual for the device, or pull up the properties for an existing XNET session in code.

Please see [TC10_Wake/Sleep](#).

Packet Focus

To focus on a packet, select a row in the table by clicking anywhere on the row. When a row is selected, the protocol layer, data layer, and payload will be displayed below the table.

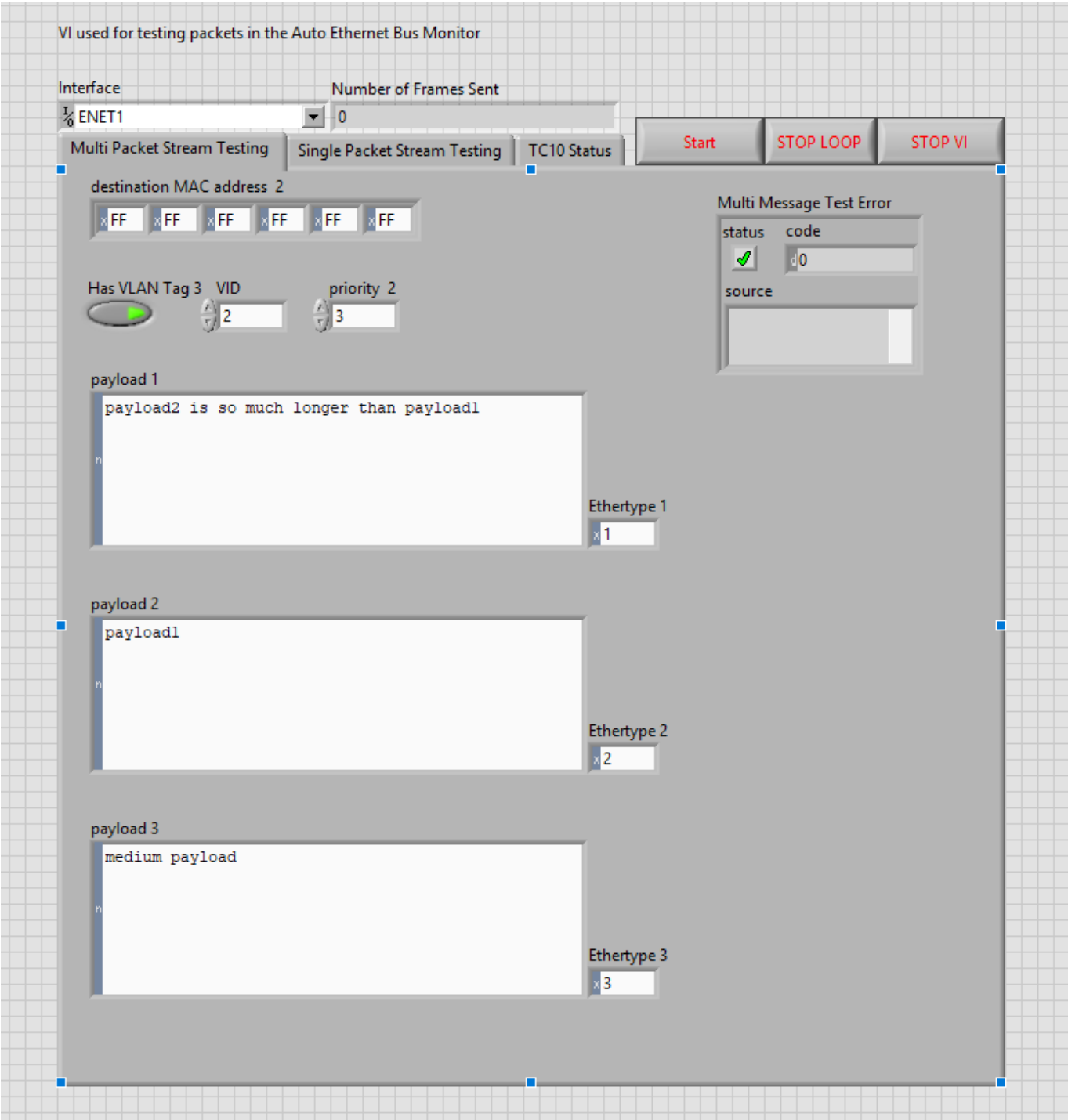
Packet_Number	Timestamp	Source	Destination	Protocol	Length	Data
275	1/6/2026 4:57:29 PM	00:80:2F:42:A0:4A	FF:FF:FF:FF:FF:FF	2	60	FF FF FF FF FF FF 00 80 2F 42 A0 4A 81 00 60 02 00 02 7
276	1/6/2026 4:57:29 PM	00:80:2F:42:A0:4A	FF:FF:FF:FF:FF:FF	1	60	FF FF FF FF FF FF 00 80 2F 42 A0 4A 81 00 60 02 00 01 7
277	1/6/2026 4:57:29 PM	00:80:2F:42:A0:4A	FF:FF:FF:FF:FF:FF	3	60	FF FF FF FF FF FF 00 80 2F 42 A0 4A 81 00 60 02 00 03 6
278	1/6/2026 4:57:30 PM	00:80:2F:42:A0:4A	FF:FF:FF:FF:FF:FF	2	60	FF FF FF FF FF FF 00 80 2F 42 A0 4A 81 00 60 02 00 02 7
279	1/6/2026 4:57:30 PM	00:80:2F:42:A0:4A	FF:FF:FF:FF:FF:FF	1	60	FF FF FF FF FF FF 00 80 2F 42 A0 4A 81 00 60 02 00 01 7
280	1/6/2026 4:57:30 PM	00:80:2F:42:A0:4A	FF:FF:FF:FF:FF:FF	3	60	FF FF FF FF FF FF 00 80 2F 42 A0 4A 81 00 60 02 00 03 6
281	1/6/2026 4:57:30 PM	00:80:2F:42:A0:4A	FF:FF:FF:FF:FF:FF	2	60	FF FF FF FF FF FF 00 80 2F 42 A0 4A 81 00 60 02 00 02 7
282	1/6/2026 4:57:31 PM	00:80:2F:42:A0:4A	FF:FF:FF:FF:FF:FF	1	60	FF FF FF FF FF FF 00 80 2F 42 A0 4A 81 00 60 02 00 01 7
283	1/6/2026 4:57:31 PM	00:80:2F:42:A0:4A	FF:FF:FF:FF:FF:FF	3	60	FF FF FF FF FF FF 00 80 2F 42 A0 4A 81 00 60 02 00 03 6
284	1/6/2026 4:57:31 PM	00:80:2F:42:A0:4A	FF:FF:FF:FF:FF:FF	2	60	FF FF FF FF FF FF 00 80 2F 42 A0 4A 81 00 60 02 00 02 7
285	1/6/2026 4:57:31 PM	00:80:2F:42:A0:4A	FF:FF:FF:FF:FF:FF	1	60	FF FF FF FF FF FF 00 80 2F 42 A0 4A 81 00 60 02 00 01 7
286	1/6/2026 4:57:32 PM	00:80:2F:42:A0:4A	FF:FF:FF:FF:FF:FF	3	60	FF FF FF FF FF FF 00 80 2F 42 A0 4A 81 00 60 02 00 03 6

SOURCE MAC ADDRESS 00:80:2F:42:A0:4A		DESTINATION MAC ADDRESS FF:FF:FF:FF:FF:FF	
VLAN TAG		ETHERTYPE 0	
TPID 33024	PRIORITY 3	DEI ☐	VID 2

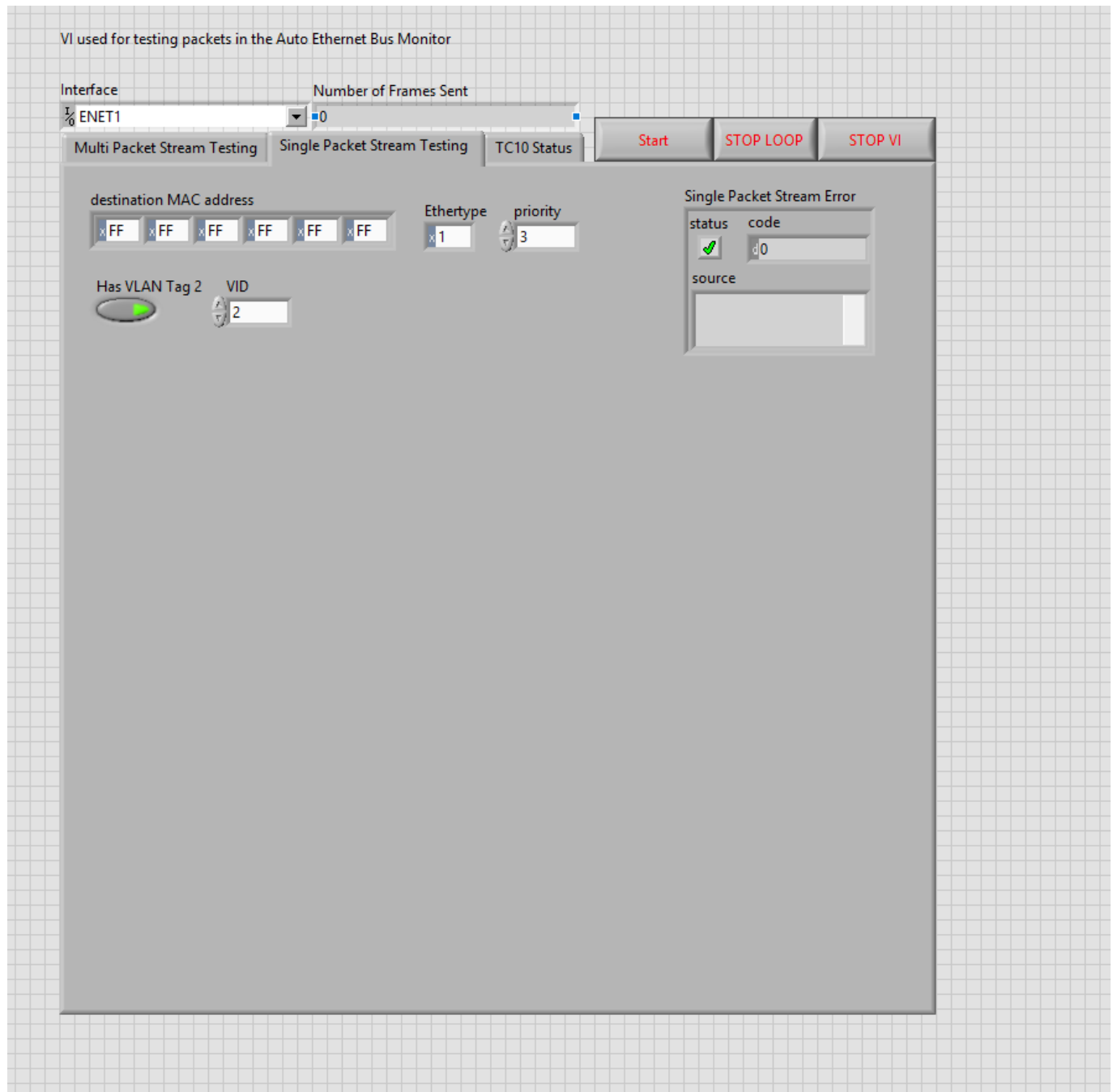
[illegible]

Testing the plugin

The first functionality is the multi Packet stream, allowing the user to send 3 different packets, varying protocol and message, sequentially.



The next functionality is the single packet stream which sends a single packet on repeat.



The last functionality reads and writes the TC10 status. This will also display if the device is TC10 Enabled.

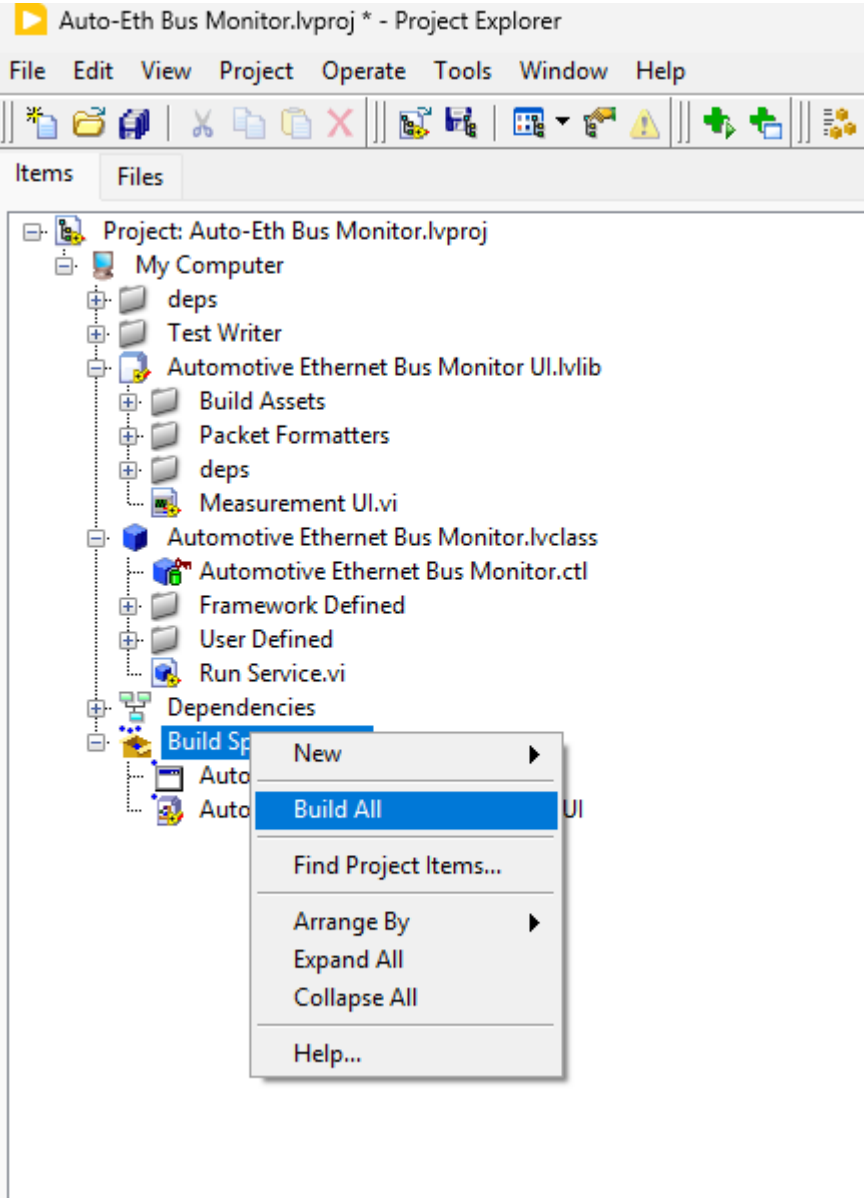
For all of the these tests, the user must first set the XNET Interface.

Troubleshooting

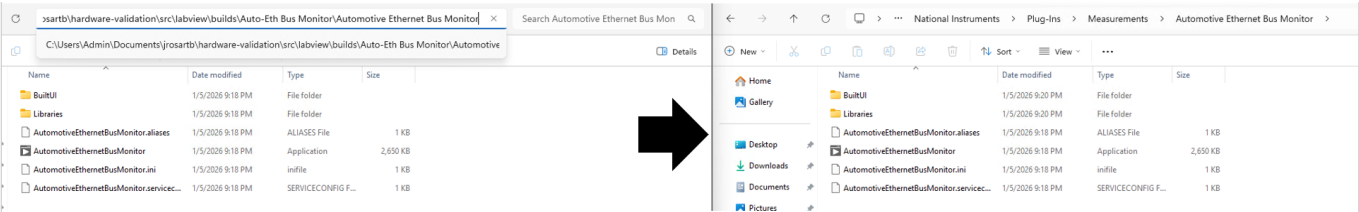
If part of the UI is getting cut off, please open the Measurement UI.vi and scroll down so the top of the screen is lined up just above the letters in the UI, then save and rebuild.

How to deploy the plugin locally

Build the **Automotive Ethernet Bus Monitor** and the **Automotive Ethernet Bus Monitor UI**



Copy the hardware-validation\src\labview\builds\Auto-Eth Bus Monitor\Automotive Ethernet Bus Monitor folder to the C:\ProgramData\National Instruments\Plug-Ins\Measurements\Automotive Ethernet Bus Monitor location.



After copying the folder above, the service will start when Instrument Studio starts, allowing the user to run the plugin independently of Labview.